RESEARCH PROPOSAL

1. Personal Details

- **Student/Applicant Names:** Phoku Gratitude Malehwiti
- **Program of Interest:** BSc Honours in Computer Science and Information Sciences
- **Proposed Research Topic:** A Comparative Evaluation of Unsupervised Anomaly Detection Algorithms for Credit Card Fraud Detection
- **Primary/Focused Research Area:** Machine Learning / Data Science
- **Application Domain:** Cybersecurity / Financial Technology (FinTech)

2. Introduction & Background

Credit card fraud brings about a multi-billion-rand threat to global financial systems annually in the world. Common traditional fraud detection systems rely heavily on rule based methods, which fail to adapt to evolving, sophisticated fraud patterns which brings about problems. While supervised machine learning models are highly accurate, they depend on historical, labelled fraud examples that are rare due to severe dataset imbalances. Consequently, unsupervised anomaly detection has emerged as a critical alternative for identifying unseen, irregular transactional behaviours without requiring prior data labelling.

3. Problem Statement

Standard classification algorithms often fail in financial fraud detection due to the extreme scarcity of fraud instances relative to legitimate transactions. Traditional systems generate high rates of false positives, which disrupts authentic user transactions and increases operational costs. There is a need to systematically evaluate how lightweight, unsupervised anomaly detection models perform against these skewed data distributions to achieve high detection rates while maintaining low operational latency.

4. Research Objectives

1. To establish a clean, optimized data pipeline using a recognized, highly imbalanced benchmark credit card transaction dataset.

2. To implement and evaluate two core unsupervised algorithms: Anomaly Detection (e.g Isolation Forest) and an optimized baseline classifier (e.g Support Vector Machine).
3. To compare the models using industry-standard metrics to determine which architecture minimizes false positives effectively.

5. Preliminary Methodology

This project follows a streamlined, structured research process:

[Data Ingestion] ➔ [Data Preprocessing] ➔ [Model Training] ➔ [Evaluation & Comparison]

- **Data Acquisition:** The study will utilize the public, anonymized European Credit Card Fraud dataset (sourced via Kaggle), removing any data collection bottlenecks or ethical compliance delays.
- **Data Preprocessing:** Handling extreme data skewness using robust scaling, dimensional reduction (PCA), and evaluating the pipeline with and without synthetic data balancing (SMOTE).
- **Model Implementation:** Algorithms will be built in Python using standard, resource-efficient libraries which we have learnt mostly in class (scikit-learn, pandas).
- **Evaluation:** Models will be explicitly measured on Classification Accuracy, Precision, Recall, and F1-score.

6. Project Feasibility

- **Timeline Feasibility:** The project scope is narrow, focusing strictly on model evaluation rather than building a full-scale production software application, ensuring completion within the academic year.
- **Resource & Compute Feasibility:** The chosen dataset fits entirely into memory, and the selected traditional machine learning architectures can execute efficiently on a standard consumer laptop or free cloud compute tiers (Google Colab).
- **Data Availability:** The dataset is fully public, structured, and immediately accessible, presenting zero risk of data access denial or POPIA compliance holds.